Speed up Spark Performance with PCIe SSD

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Agenda

- PCIe SSD Overview
- Use PCIe SSD to accelerate computing
- Why SSD is necessary in Spark

PCIe SSD Overview

- 突破性的性能

面向 PCIe* 的英特尔® 固态盘数据中心家族具备多达 2.8GB/秒和 460K IOPS（每秒输入/输出操作）的数据读取性能，以及多达 2.0GB/秒和 175K IOPS 的数据写入性能。

面向 PCIe* 的英特尔® 固态盘数据中心家族具备极其卓越的性能，只需一块硬盘便能够取代七块通过 HBA 聚合在一起的 SATA 固态盘（约 500K IOPS），其中尤以英特尔® 固态盘数据中心 P3700 的性能最为出色 (460K IOPS)。对于性能为 200 IOPS 的传统机械硬盘而言，需要组合使用 2,300 块 15K 机械硬盘，才能媲美一块面向 PCIe* 的英特尔® 固态盘数据中心家族。

- 插卡式（AIC）

- 2.5 英寸 U.2 (SFF8369 支持热插拔)

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- Why SSD is necessary in Spark

Use PCIe SSD to accelerate computing - Motivation

- Usually customers' servers have HDDs (7-11 usually) already,

  so we propose to add ***only 1 PCIe SSD*** as ***cache*** for hot data and HDDs as backup storage.

Use PCIe SSD to accelerate computing - Motivation

- Usually customers' servers have HDDs (7-11 usually) already,

  so we propose to add ***only 1 PCIe SSD*** as ***cache*** for hot data and HDDs as backup storage.
- Tachyon is an existing solution, but:
  - ❑ Only supporting RDD cache, not including shuffle data
  - ❑ Extra software component, extra deployment and maintain effort
  - ❑ Extra performance loss to run tachyon daemon and inter-process communication

Use PCIe SSD to accelerate computing - Implementation

[SPARK-12196][Core] Store/retrieve blocks in different speed storage devices by hierarchy way #10225

Open **yucai** wants to merge 6 commits into `apache:master` from `yucai:hs`

- When Spark core allocates files (either for RDD cache or shuffle), it gets files from PCIe SSD first, after PCIe SSD's useable space is less than some threshold, getting files from HDDs.
- Yarn dynamical allocation is supported also.

Use PCIe SSD to accelerate computing - Usage

1. Set the priority and threshold in spark-default.xml.

```
spark.storage.hierarchyStore    ssd 30GB
```

2. Configure “ssd” location: just put the keyword like "ssd“ in local dir.
   For example, in yarn-site.xml:

```
<property>
  <name>yarn.nodemanager.local-dirs</name>
  <value>file:///mnt/DP_disk1/yucai/yarn/local,file:///mnt/DP_disk2/yucai/yarn/local,
         file:///mnt/DP_disk3/yucai/yarn/local,file:///mnt/DP_disk4/yucai/yarn/local,
         file:///mnt/DP_disk5/yucai/yarn/local,file:///mnt/DP_disk6/yucai/yarn/local,
         file:///mnt/DP_disk7/yucai/yarn/local,file:///mnt/DP_disk8/yucai/yarn/local,
         file:///mnt/DP_disk9/yucai/yarn/local,file:///mnt/DP_disk10/yucai/yarn/local,
         file:///mnt/DP_disk11/yucai/yarn/local, file:///mnt/nvme0n1/yucai/yarn.local.ssd,
  </value>
</property>
```

Real world Spark adoptions

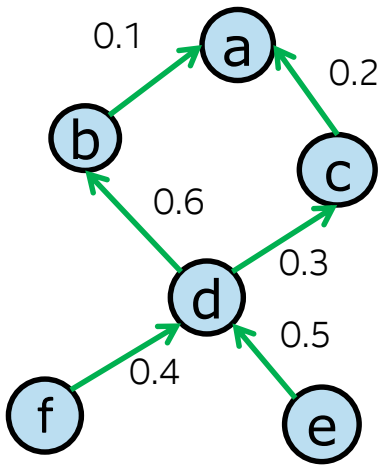
Workload	Category	Description	Rationale	Customer
NWeight	Machine Learning	To compute associations between two vertices that are *n-hop* away(e.g., friend–to-friend associations or similarities between videos for recommendation).	Iterative graph-parallel algorithm, implemented with Bagel (Pregel on Spark) and/or GraphX (new Graph parallel framework on Spark)	Youku
SparkSQL	SQL	Real analysis queries from Baidu NuoMi.	Left Outer Join, Group BY, UNION etc.	Baidu

NWeight Introduction

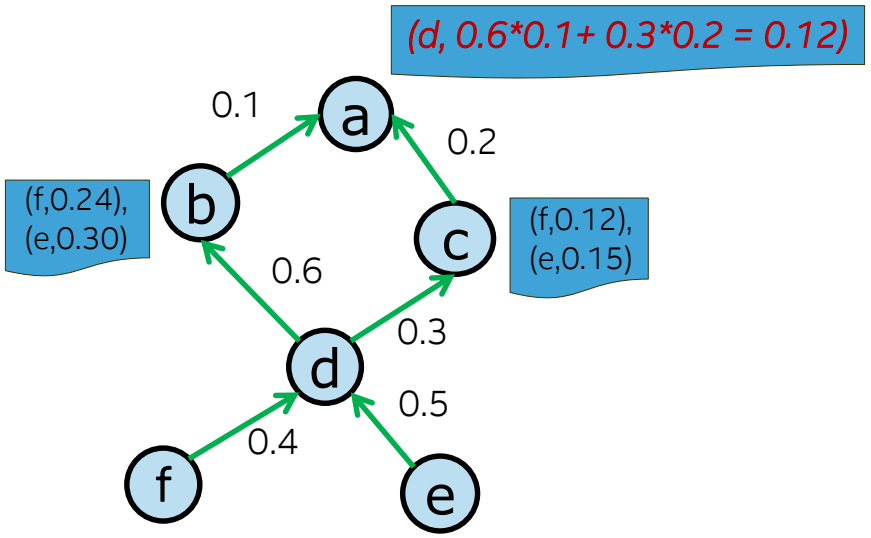
To compute associations between two vertices that are n-hop away.

e.g., friend –to-friend, or similarities between videos for recommendation

Initial directed graph

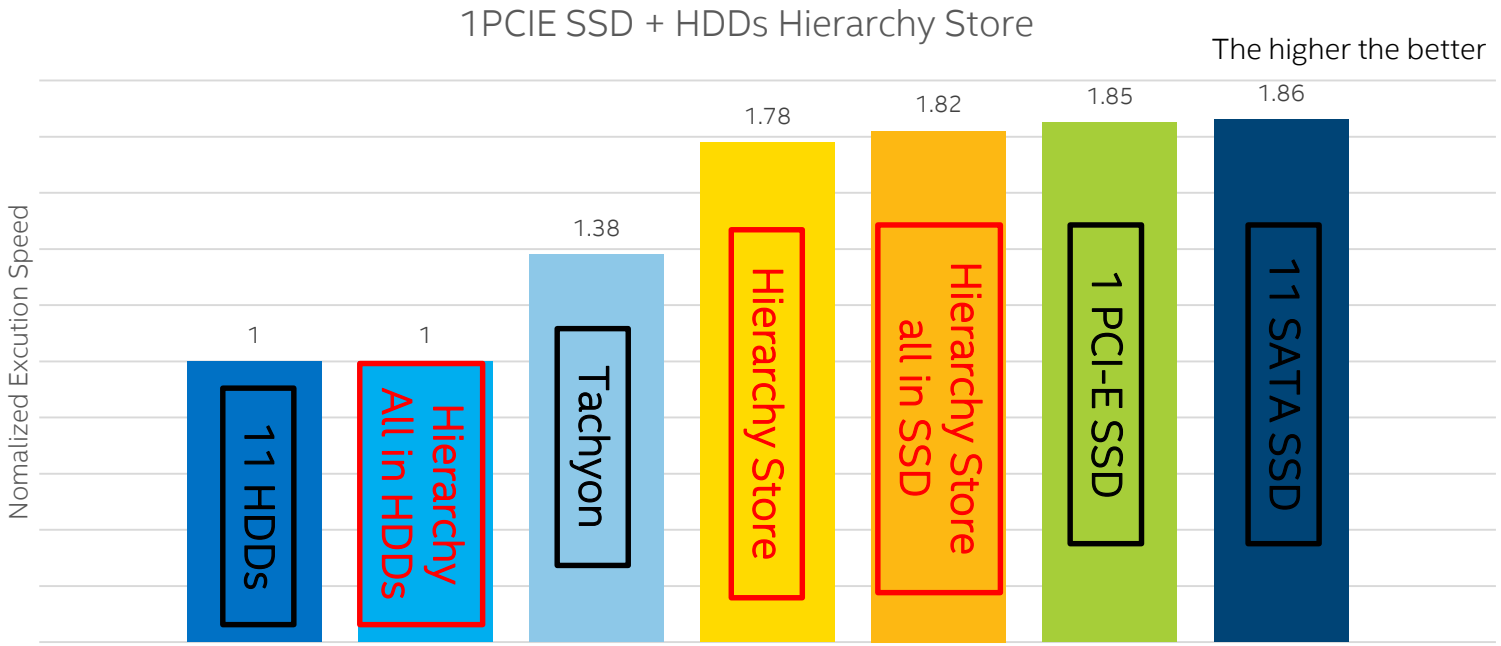


2-hop association



PCIe SSD hierarchy store performance report

- Pure SSD scenario: 1 PCIe SSD performs the same as 11 SATA SSDs (SSD shifts bottleneck to CPU).
- For our hierarchy store solution:
  - ✓ No extra overhead: best case the same with pure SSD (PCIe/SATA SSD), worst case the same with pure HDDs.
  - ✓ Compared with 11 HDDs, *x1.82* improvement at least (CPU limitation).
  - ✓ Compared with Tachyon, still shows *x1.3* performance advantage: cache both RDD and shuffle, no inter-process communication.



An Analysis SQL from Baidu NuoMi

Run 8 instance simultaneously

Stage Id	Tasks	Input	Shuffle Read	Shuffle Write
22	42685	61.1 KB		4.8 KB
20	26/26	3.1 GB		1982.7 MB
19	62/62	30.3 MB		15.2 MB
16	42685	61.1 KB		4.8 KB
14	26/26	3.1 GB		1982.7 MB
12	497/497	56.9 GB		36.0 GB
11	100/100	178.2 MB		77.6 MB
8	42685	61.1 KB		4.8 KB
6	26/26	3.1 GB		1982.7 MB
5	273/273	203.6 MB		145.7 MB

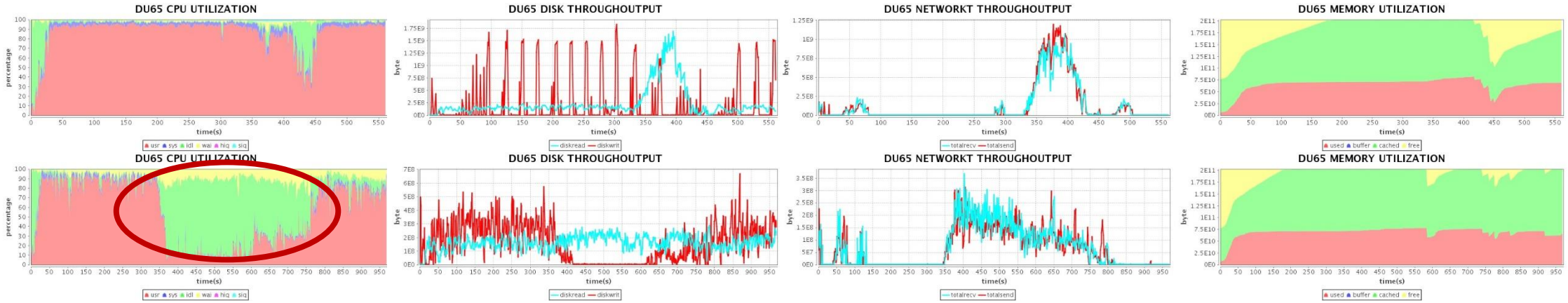
Stage Id	Tasks	Input	Shuffle Read	Shuffle Write
24	320/320		15.8 MB	
23	960/960		88.2 MB	15.8 MB
18	320/320		32.5 MB	45.1 MB
21	320/320		1997.9 MB	3.5 MB
17	320/320		16.9 MB	32.5 MB
15	320/320		2.1 GB	16.9 MB
13	320/320		36.0 GB	206.3 MB
10	320/320		26.7 MB	39.9 MB
9	320/320		16.6 MB	26.7 MB
7	320/320		2.1 GB	16.6 MB

SSD v.s. HDDs: x1.7 end-to-end improvement

x1.7 end-to-end improvement (7 mins v.s. 12 mins)
x5 shuffle improvement (1 min v.s. 5 mins)

x6 Disk BW

x4 network BW



Significant IO Wait

Device:	rrqm/s	wrqm/s	r/s	w/s	rkB/s	wkB/s	avgrq-sz	avgqu-sz	await	svctm	%util
nvme0n1	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
sda	0.00	0.00	0.00	9.00	0.00	36.00	8.00	0.00	0.00	0.00	0.00
sdc	2.00	0.00	398.00	0.00	37268.00	0.00	187.28	7.09	15.63	2.46	97.90
sdb	19.00	347.00	389.00	8.00	35236.00	1420.00	184.66	10.37	25.28	2.45	97.30
sdd	19.00	0.00	326.00	23.00	29592.00	11776.00	237.07	153.35	137.34	2.87	100.00
sde	0.00	0.00	317.00	0.00	30344.00	0.00	191.44	2.87	9.02	2.54	80.40
sdf	11.00	0.00	267.00	1.00	25332.00	4.00	189.07	12.98	50.60	3.73	100.00
sdg	18.00	332.00	384.00	0.00	34684.00	0.00	180.65	25.56	47.58	2.60	100.00
sdh	4.00	183.00	334.00	5.00	33392.00	752.00	201.44	4.86	14.35	2.69	91.30

Significant IO bottleneck

Agenda

- PCIe SSD Overview
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- Why SSD is necessary in Spark

Deep dive into the real customer case

NWeight

x3 improvement!!

						11 HDDs	PCIe SSD
Stage Id	Description	Input	Output	Shuffle Read	Shuffle Write	Duration	Duration
23	saveAsTextFile at BagelNWeight.scala:102+details	50.1 GB	27.6 GB			27 s	20 s
17	foreach at Bagel.scala:256+details	732.0 GB		490.4 GB		*23 min*	*7.5 min*
16	flatMap at Bagel.scala:96+details	732.0 GB			490.4 GB	15 min	13 min
11	foreach at Bagel.scala:256+details	590.2 GB		379.5 GB		*25 min*	11 min
10	flatMap at Bagel.scala:96+details	590.2 GB			379.6 GB	12 min	10 min
6	foreach at Bagel.scala:256+details	56.1 GB		19.1 GB		4.9 min	3.7 min
5	flatMap at Bagel.scala:96+details	56.1 GB			19.1 GB	1.5 min	1.5 min
2	foreach at Bagel.scala:256+details			15.3 GB		38 s	39 s
1	parallelize at BagelNWeight.scala:97+details					38 s	38 s
0	flatMap at BagelNWeight.scala:72+details	22.6 GB			15.3 GB	46 s	46 s

5 Main IO pattern

	Map Stage	Reduce Stage
RDD	rdd_read_in_map	rdd_read_in_reduce rdd_write_in_reduce
Shuffle	shuffle_write_in_map	shuffle_read_in_reduce

How to do IO characterization?

- We use blktrace* to monitor each IO to disk. Such as:

```
8,32   5    13561    43.740003038 13615  D   W 52090704 + 560 [java]
8,32   5    13567    43.741721414 13615  D   R 13637888 + 256 [java]
8,32   5    13569    43.741778543 13615  C   R 13637888 + 256 [0]
8,32   5    13570    43.743476793 13615  C   W 52089584 + 560 [0]
8,32   5    13571    43.743519464 13615  D   W 52091264 + 560 [java]
8,32   5    13577    43.743901725 13615  D   R 13638144 + 256 [java]
8,32   5    13579    43.745785309 13615  C   W 52090144 + 560 [0]
8,32   5    13585    43.746306216 13615  D   R 13638400 + 256 [java]
8,32   5    13587    43.748340765     0  C   W 52090704 + 560 [0]
8,32  17     5938    43.761506738 13449  D   R 50251520 + 8 [java]
```

Start to write 560 sectors from address 52090704

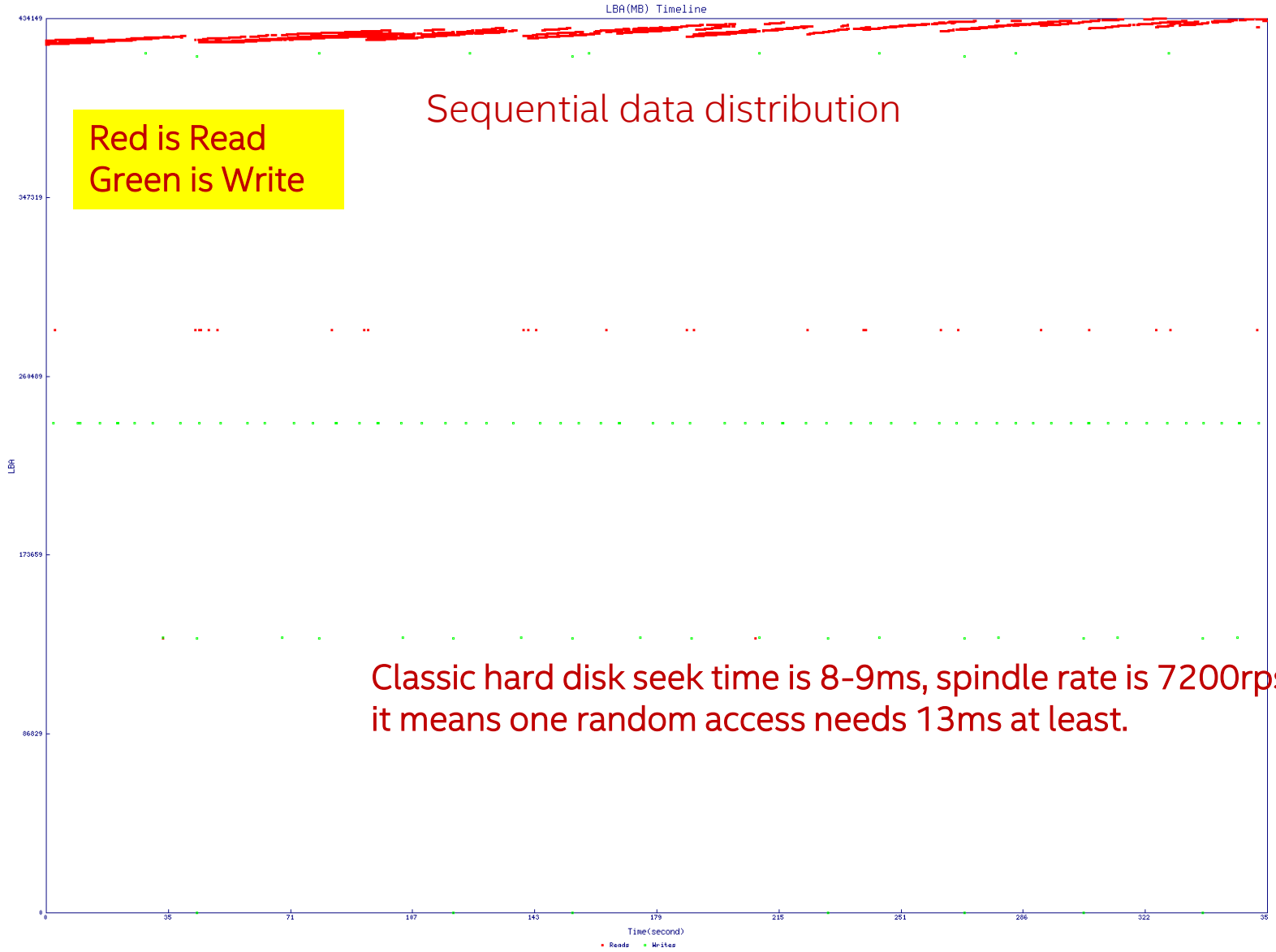
Start to read 256 sectors from address 13637888
Finish the previous read command (13637888 + 256)

Finish the previous write command (52090704 + 560)

- We parse those raw info, generating 4 kinds of charts: IO size histogram, latency histogram, seek distance histogram and LBA timeline, from which we can identify the IO is sequential or random.

* blktrace is a kernel block layer IO tracing mechanism which provides detailed information about disk request queue operations up to user space.

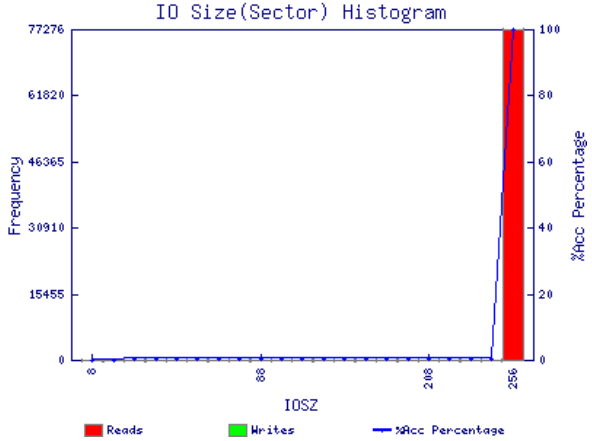
RDD Read in Map: sequential



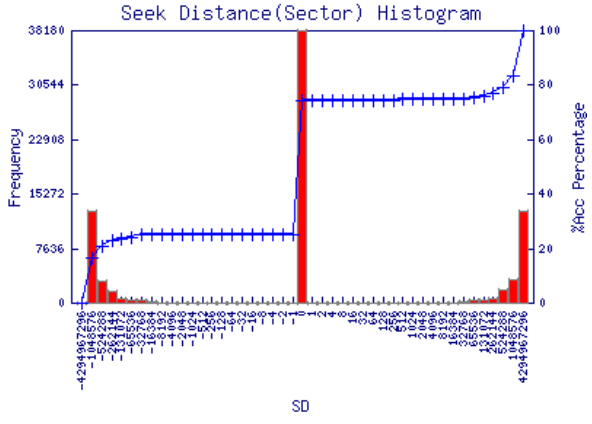
Red is Read
Green is Write

Sequential data distribution

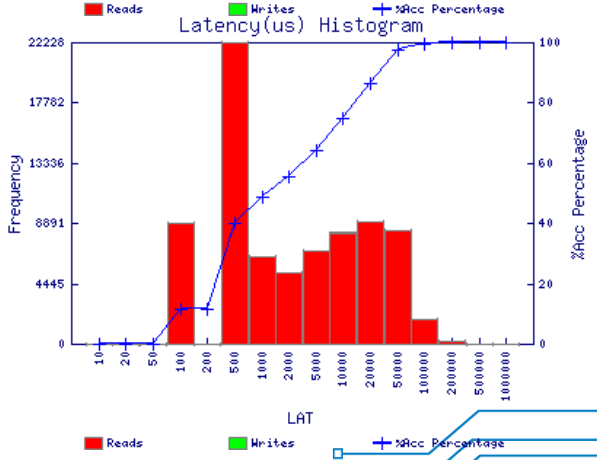
Classic hard disk seek time is 8-9ms, spindle rate is 7200rps, it means one random access needs 13ms at least.



Big IO size

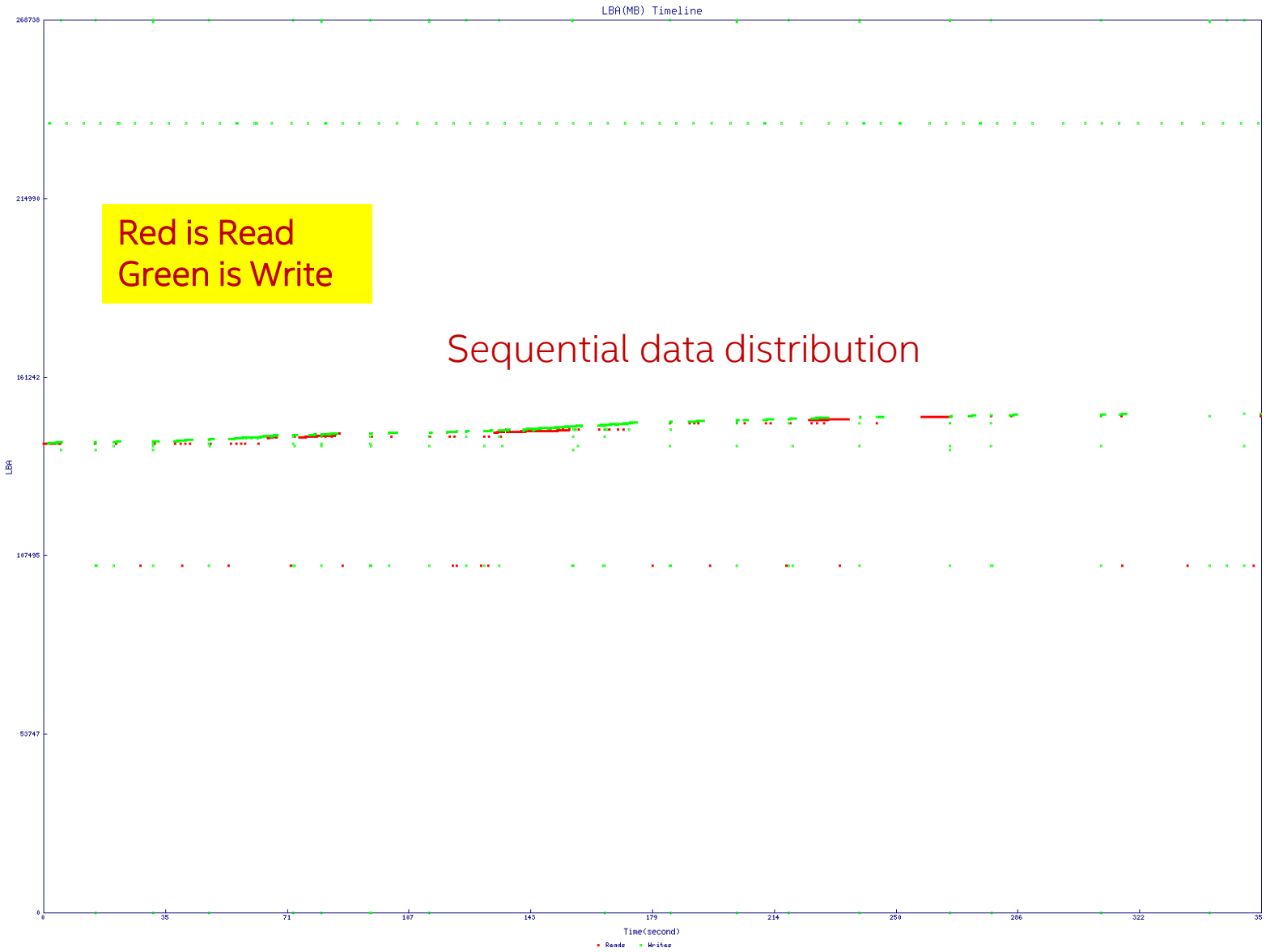


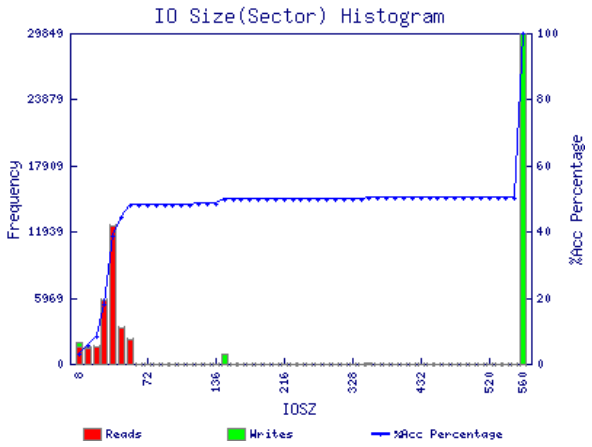
Much 0 SD



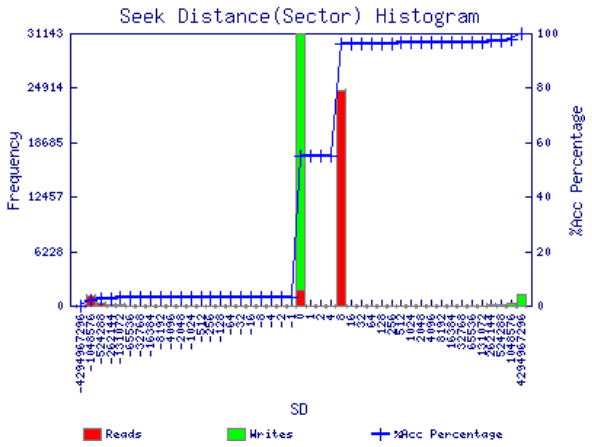
Low latency

Shuffle Write in Map: sequential



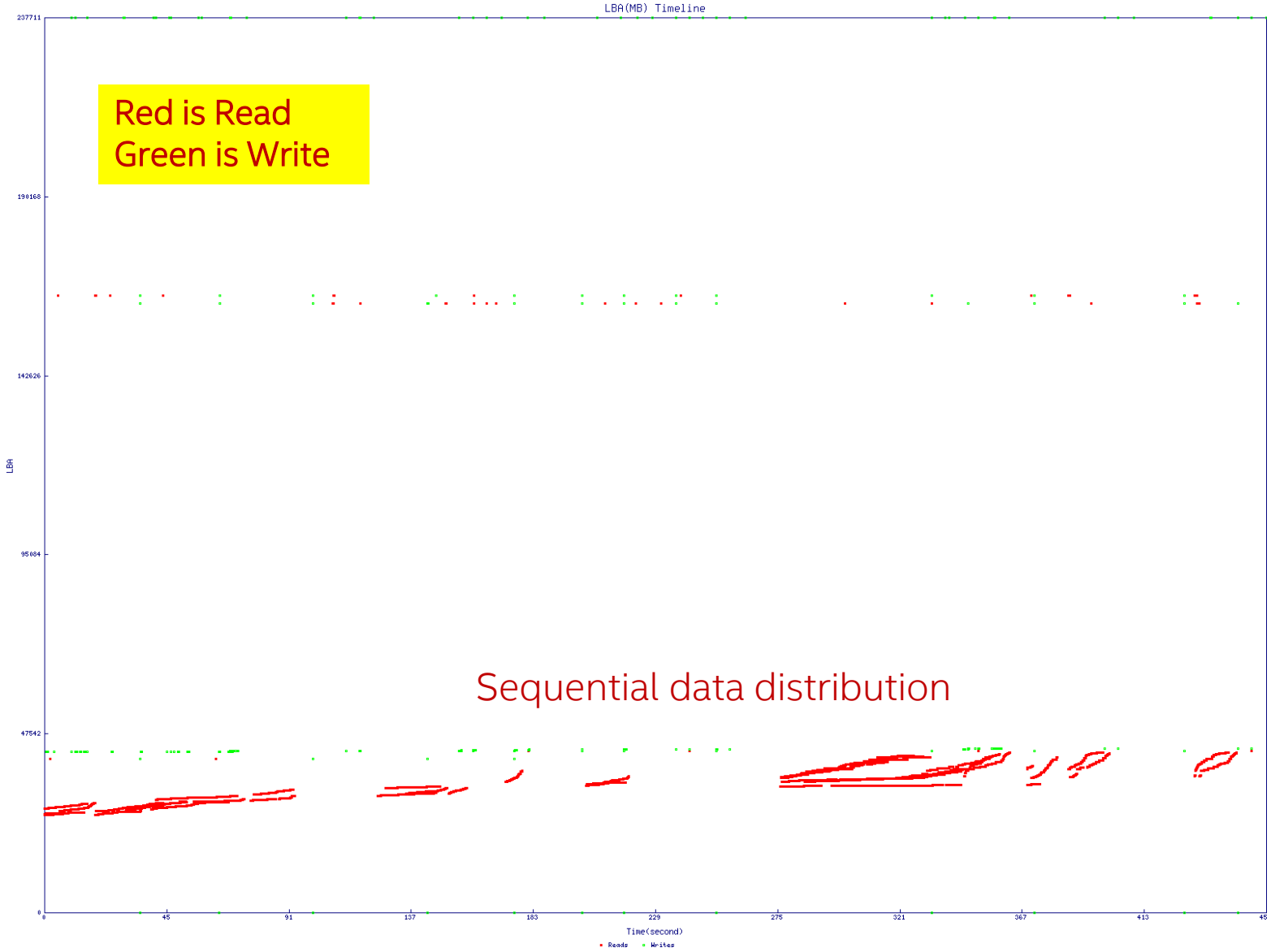


Big IO size



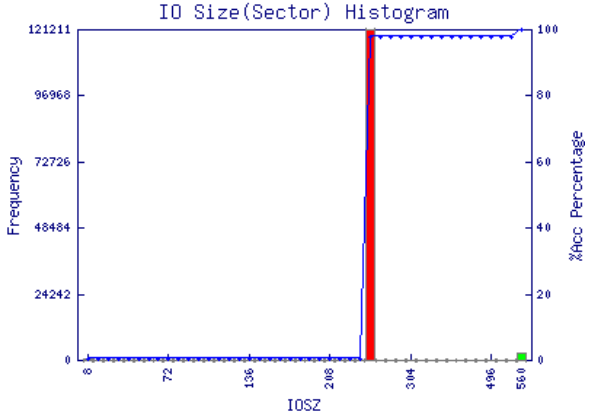
Much 0 SD

RDD Read in Reduce: sequential

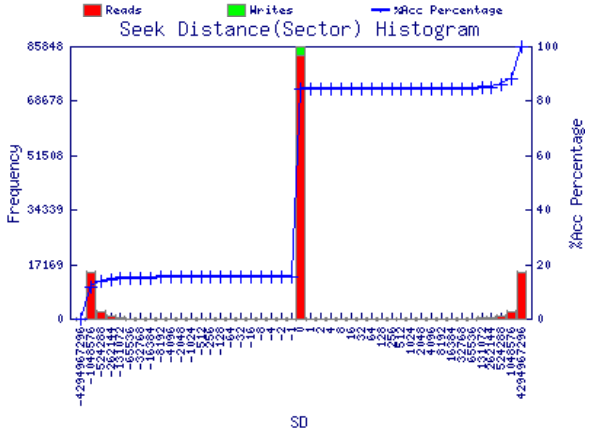


Red is Read
Green is Write

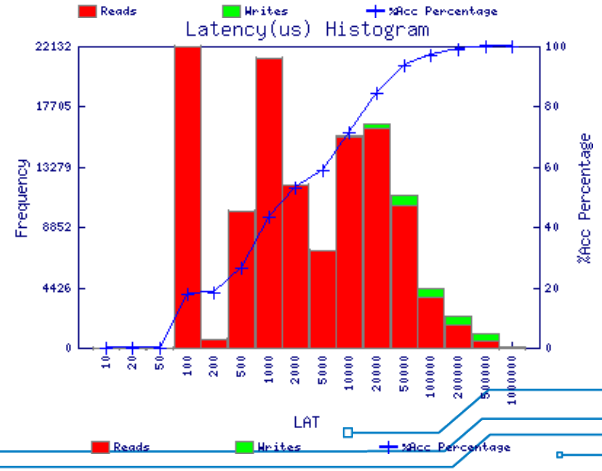
Sequential data distribution



Big IO size

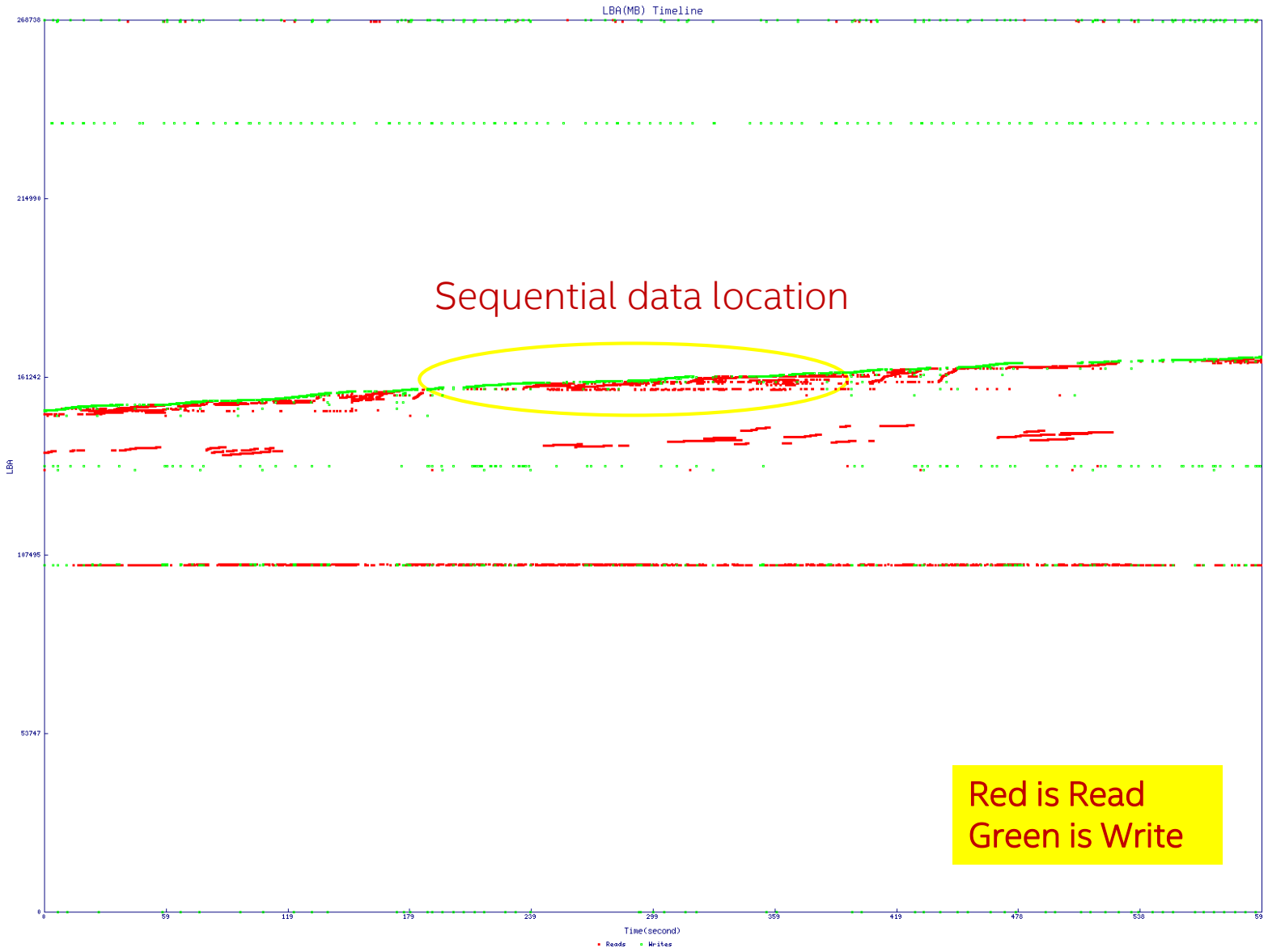


Much 0 SD

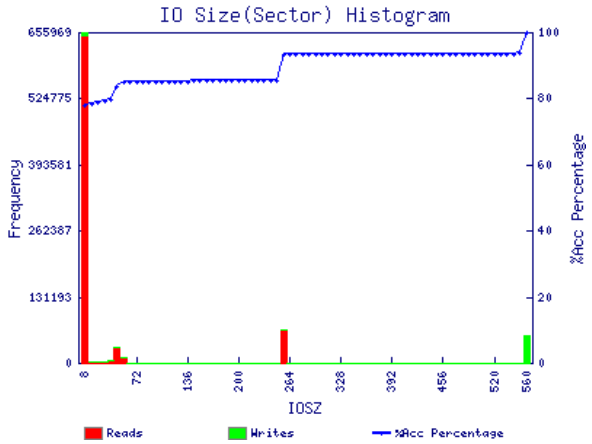


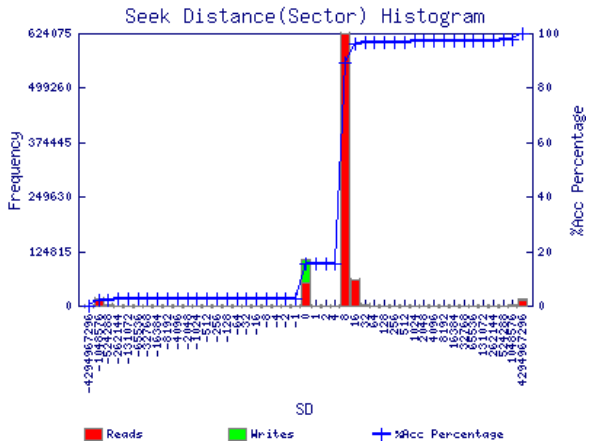
Low latency

RDD Write in Reduce: sequential write but with frequent 4K read



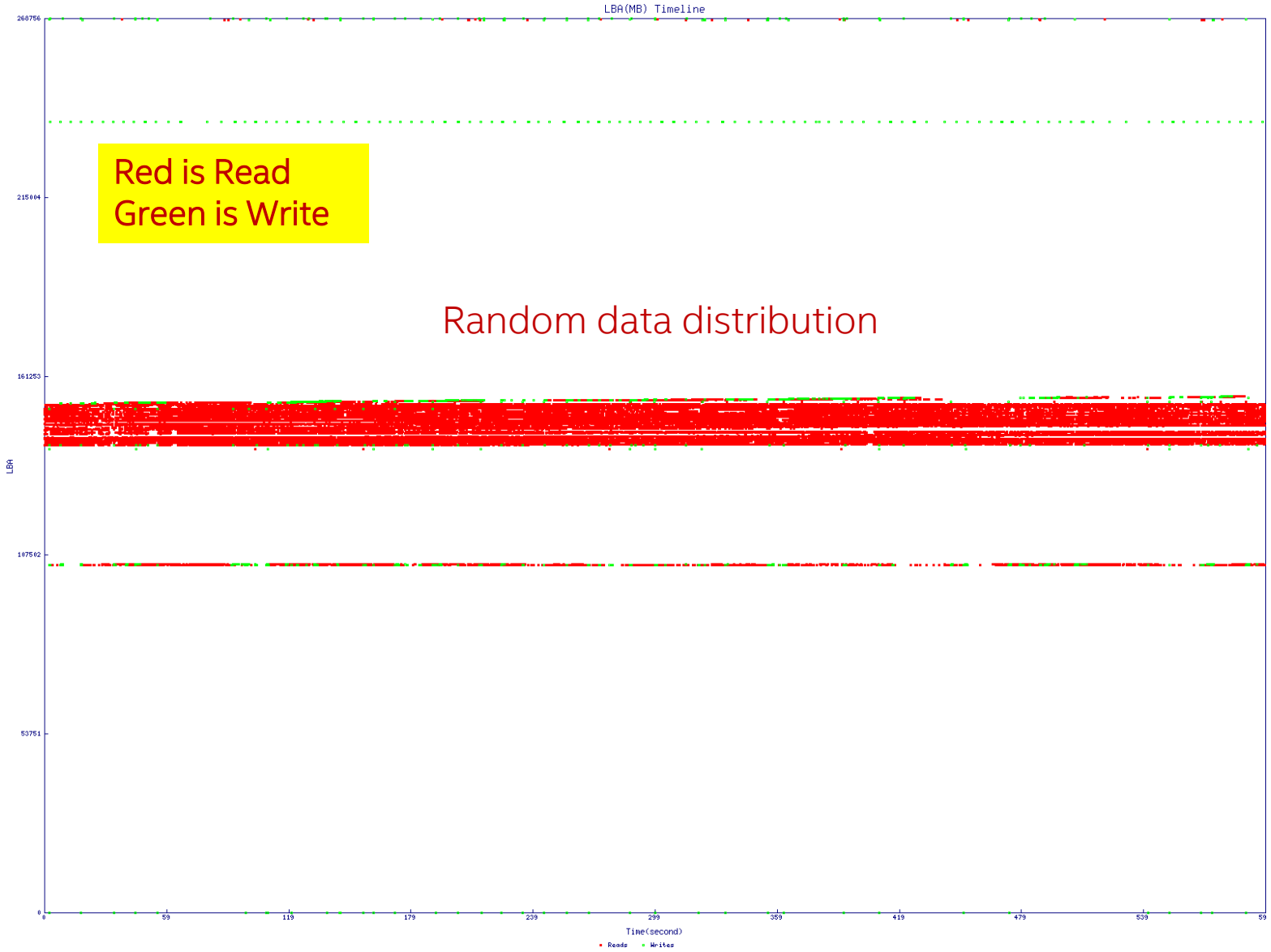
Those 4K read is probably because of spilling in cogroup, maybe a spark issue





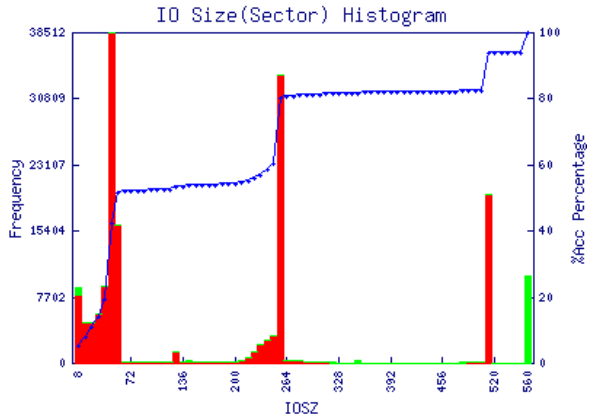
Write IO size is big but with many small 4K read IO

Shuffle Read in Reduce: random

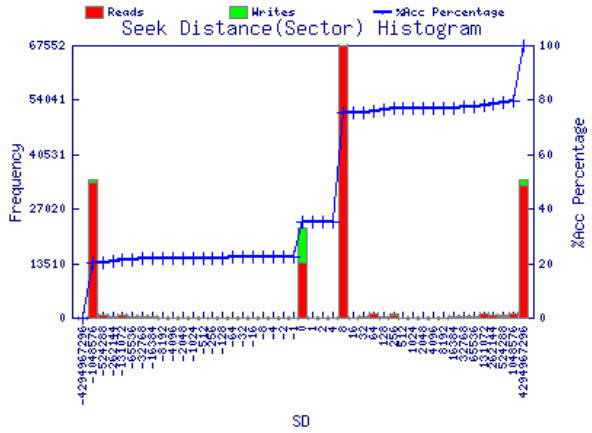


Red is Read
Green is Write

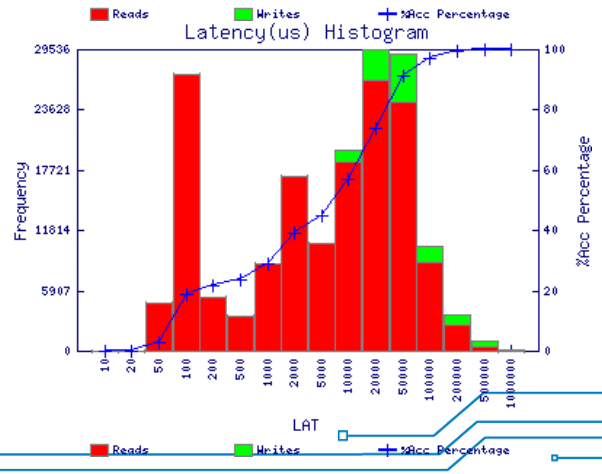
Random data distribution



Small IO size

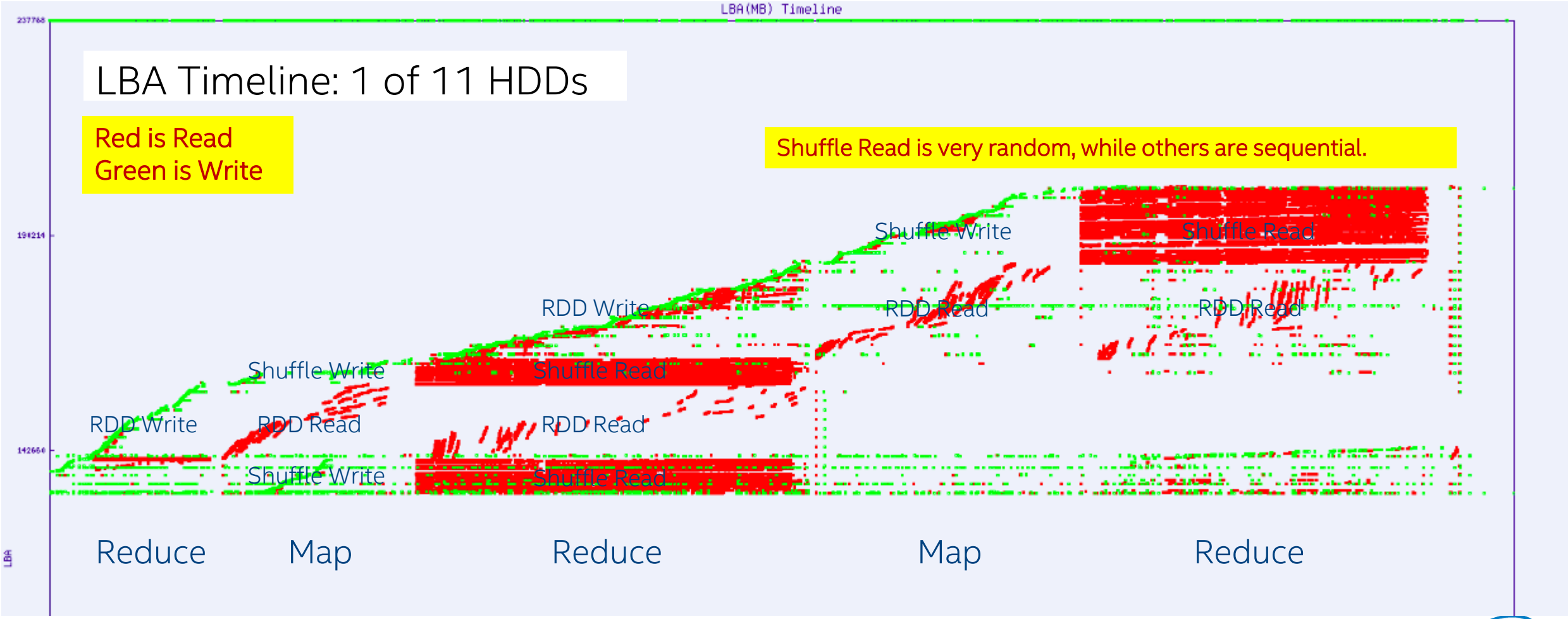


Few 0 SD



High latency

Overall Disk IO Picture



Conclusion

- RDD read/write, shuffle write are sequential.
- Shuffle read is random.

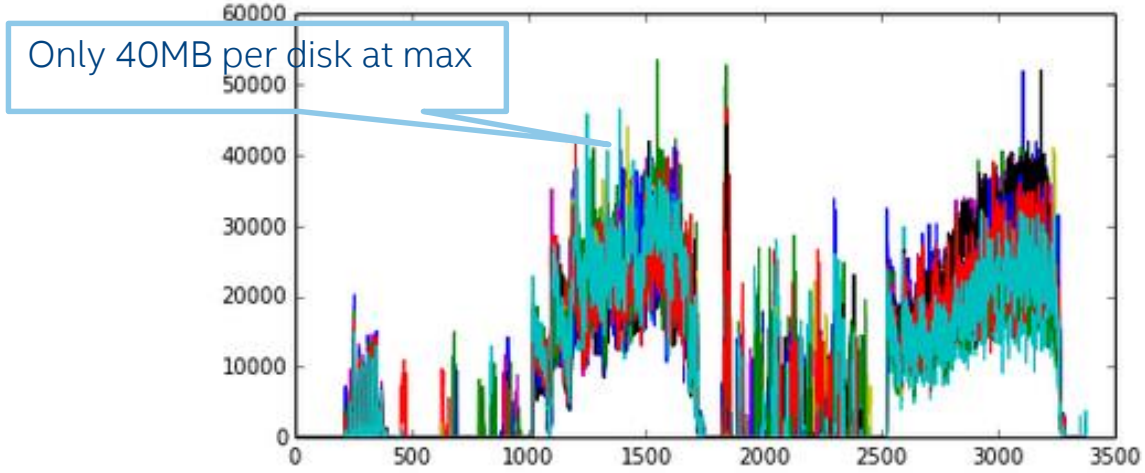
Type	IO Characterization
rdd_read_in_map	Sequential
shuffle_write_in_map	
rdd_read_in_reduce	
rdd_write_in_reduce	
shuffle_read_in_reduce	Random

Using SSD to speed up “shuffle read in reduce”

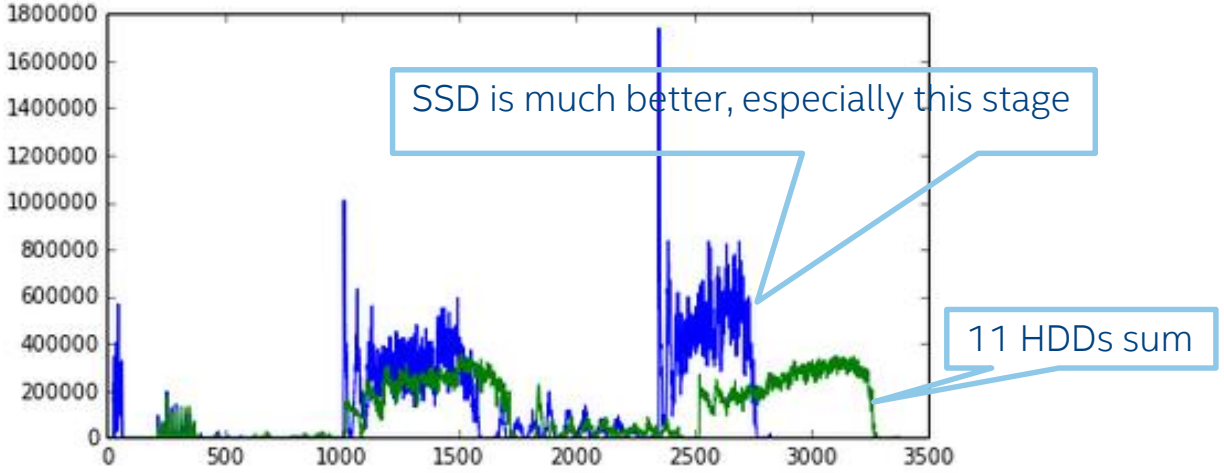
x2 improvement for “shuffle read in reduce”
x3 improvement in real shuffle
x2 improvement in E2E testing

CPU is still the bottleneck !

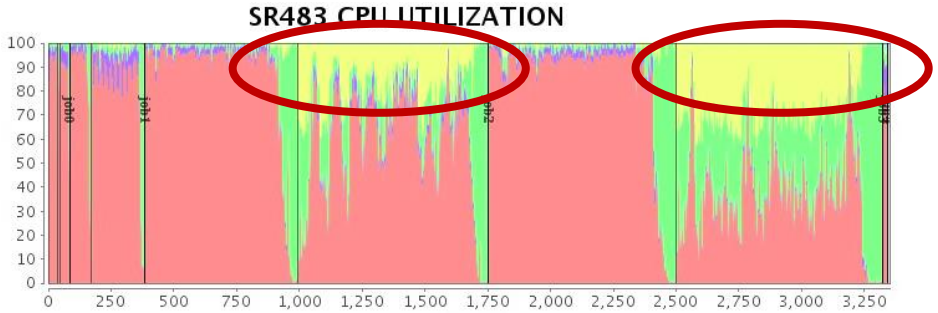
Per disk BW when shuffle read from HDD



BW when shuffle read from SSD

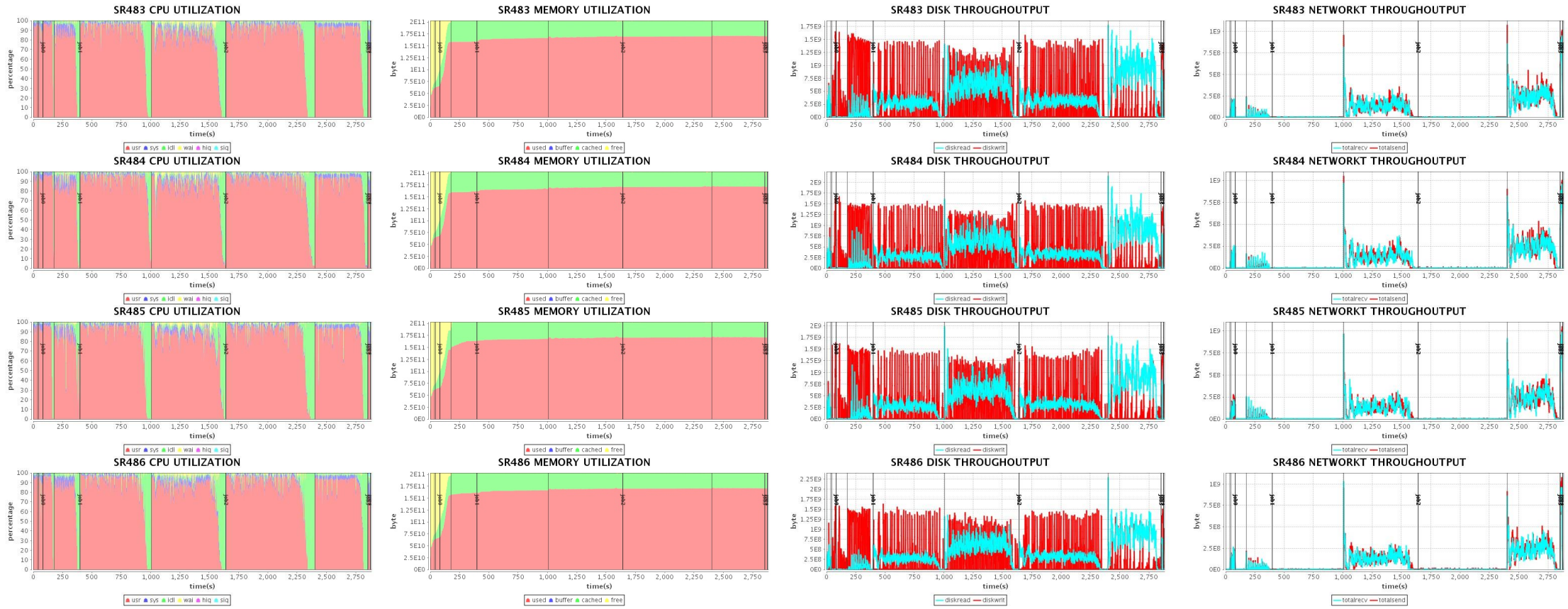


Shuffle read from HDD leads to High IO Wait



Description	Shuffle Read	Shuffle Write	SSD-RDD + HDD-Shuffle	1 SSD
saveAsTextFile at BagelNWeight.scala			20 s	20 s
foreach at Bagel.scala	490.3 GB		14 min	7.5 min
flatMap at Bagel.scala		490.4 GB	12 min	13 min
foreach at Bagel.scala	379.5 GB		13 min	11 min
flatMap at Bagel.scala		379.6 GB	10 min	10 min
foreach at Bagel.scala	19.1 GB		3.5 min	3.7 min
flatMap at Bagel.scala		19.1 GB	1.5 min	1.5 min
foreach at Bagel.scala	15.3 GB		38 s	39 s
parallelize at BagelNWeight.scala			38 s	38 s
flatMap at BagelNWeight.scala		15.3 GB	46 s	46 s

PCIe SSD Hierarchy kills IO bottleneck



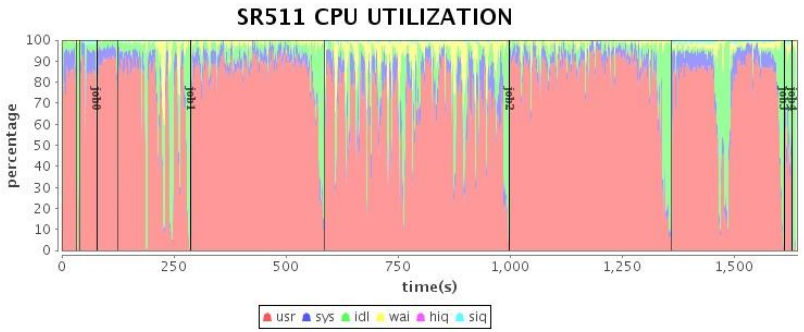
Software

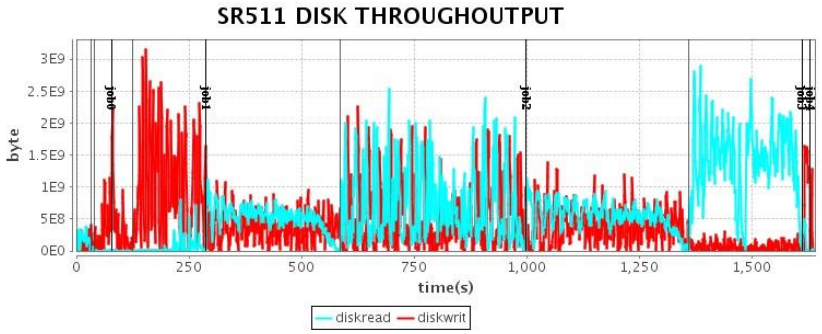
If CPU is not bottleneck?

x3-5 improvement for shuffle
x2 improvement for map stage
x3 improvement in E2E testing

NWeight

						11 HDDs	PCIe SSD	HSW
Stage Id	Description	Input	Output	Shuffle Read	Shuffle Write	Duration	Duration	Duration
23	saveAsTextFile at BagelNWeight.scala:102+details	50.1 GB	27.6 GB			27 s	20 s	26 s
17	foreach at Bagel.scala:256+details	732.0 GB		490.4 GB		*23 min*	*7.5 min*	*4.6 min*
16	flatMap at Bagel.scala:96+details	732.0 GB			490.4 GB	15 min	13 min	6.3 min
11	foreach at Bagel.scala:256+details	590.2 GB		379.5 GB		*25 min*	*11 min*	*7.1 min*
10	flatMap at Bagel.scala:96+details	590.2 GB			379.6 GB	12 min	10 min	5.3 min
6	foreach at Bagel.scala:256+details	56.1 GB		19.1 GB		4.9 min	3.7 min	2.8 min
5	flatMap at Bagel.scala:96+details	56.1 GB			19.1 GB	1.5 min	1.5 min	47 s
2	foreach at Bagel.scala:256+details			15.3 GB		38 s	39 s	36 s
1	parallelize at BagelNWeight.scala:97+details					38 s	38 s	35 s
0	flatMap at BagelNWeight.scala:72+details	22.6 GB			15.3 GB	46 s	46 s	43 s





BACKUP

SUT #A

		IVB
Master	CPU	Intel(R) Xeon(R) CPU E5-2680 @ 2.70GHz (16 cores)
	Memory	64G
	Disk	2 SSD
	Network	1 Gigabit Ethernet
Slaves	Nodes	4
	CPU	Intel(R) Xeon(R) CPU E5-2680 v2 @ 2.80GHz (2 CPUs, 10 cores, 40 threads)
	Memory	192G DDR3 1600MHz
	Disk	11 HDDs/11 SSDs/1 PCI-E SSD(P3600)
	Network	10 Gigabit Ethernet

	IVB E5-2680
OS	Red Hat 6.2
Kernel	3.16.7-upstream
Spark	Spark 1.4.1
Hadoop/HDFS	Hadoop-2.5.0-cdh5.3.2
JDK	Sun hotspot JDK 1.8.0 (64bits)
Scala	scala-2.10.4

SUT #B

		HSW
Master	CPU	Intel(R) Xeon(R) CPU X5570 @ 2.93GHz (16 cores)
	Memory	48G
	Disk	2 SSD
	Network	1 Gigabit Ethernet
Slaves	Nodes	4
	CPU	Intel(R) Xeon(R) CPU E5-2699 v3 @ 2.30GHz (2 CPUs, 18 cores, 72 threads)
	Memory	256G DDR4 2133MHz
	Disk	11 SSD
	Network	10 Gigabit Ethernet

	HSW E5-2699
OS	Ubuntu 14.04.2 LTS
Kernel	3.16.0-30-generic.x86_64
Spark	Spark 1.4.1
Hadoop/HDFS	Hadoop-2.5.0-cdh5.3.2
JDK	Sun hotspot JDK 1.8.0 (64bits)
Scala	scala-2.10.4

SUT #C

Hardware Specification		
Slaves	Nodes	4
	CPU	Intel(R) Xeon(R) CPU E5-2650 v3 @ 2.30GHz (40 cores)
	Memory	192GB/128GB DDR4 2133MHz
	Disk	7 HDDs + 1 PCIE-SSD (P3700)
	Network	10 Gigabit Ethernet
Master	CPU	Intel(R) Xeon(R) CPU E5-2640 v3 @ 2.60GHz (32 cores)
	Memory	64G
	Disk	2 SSD
	Network	10 Gigabit Ethernet

HasWell E5-2650 Software Specification	
OS	CentOS release 6.3 (Final)
Kernel	Baidu 2.6.32_1-18-0-0
Spark	Spark 1.6.0 with SSD hierarch support
Hadoop/HDFS	Hadoop-2.5.0-cdh5.3.2
JDK	Sun hotspot JDK 1.8.0 (64bits)
Scala	scala-2.10.4

Test Configuration

executors number: 32

executor memory: 18G

executor-cores: 5

spark-defaults.conf:

spark.serializer	org.apache.spark.serializer.KryoSerializer
spark.kryo.referenceTracking	false

HDD (Seagate ST9500620NS) SPEC

General

Device Type	Hard drive - internal
Capacity	500 GB
Form Factor	2.5"
Interface	Serial ATA-600
Buffer Size	64 MB
Bytes per Sector	512
Features	Native Command Queuing (NCQ), Perpendicular Magnetic Recording (PMR), PowerChoice , S.M.A.R.T.
Width	2.8 in
Depth	4 in
Height	0.6 in
Weight	6.5 oz
First Seen On Google Shopping	December 2010

HDD (Seagate ST9500620NS) SPEC

Performance

Drive Transfer Rate	600 MBps (external)
Seek Time	8.5 ms (average)
Track-to-Track Seek Time	0.7 ms
Average Latency	4.16 ms
Spindle Speed	7200 rpm

Reliability

MTBF	1,400,000 hours
Non-Recoverable Errors	1 per 10^15

PCIe SSD(P3600) SPEC

Table 6: Sequential Read and Write Bandwidth

Specification	Unit	Intel SSD DC P3600 Series				
		400GB	800GB	1.2TB	1.6TB	2TB
Sequential Read (up to)[1]	MB/s	2,100	2,600	2,600	2,600	2,600
Sequential Write (up to)[1]	MB/s	550	1000	1250	1600	1700

NOTE: Performance measured using IOMeter* with 128 KB (131,072 bytes) of transfer size with Queue Depth 128. Power mode set at 25W.

PCIe SSD(P3600) SPEC

Table 4: Random Read/Write Input/Output Operations Per Second (IOPS)

Specification[1]	Unit	Intel SSD DC P3600 Series				
		400GB	800GB	1.2TB	1.6TB	2TB
Random 4KB 70/30 Read/Write (up to)[2]	IOPS	80,000	110,000	130,000	160,000	160,000
Random 8KB 70/30 Read/Write (up to)[2]	IOPS	45,000	55,000	65,000	75,000	80,000
Random 4KB Read (up to)	IOPS	320,000	430,000	450,000	450,000	450,000
Random 4KB Write (up to)	IOPS	30,000	50,000	50,000	56,000	56,000
Random 8KB Read (up to)	IOPS	180,000	250,000	260,000	270,000	275,000
Random 8KB Write (up to)	IOPS	19,000	26,000	27,000	33,000	33,000

NOTES:

1. Performance measured using IOMeter* on Intel provided Windows Server 2012 R2 driver with Queue Depth 32 and number of workers equal to 4. Measurements are performed on a full Logical Block Address (LBA) span of the drive. Power mode set at 25W.
2. 4KB = 4,096 bytes
3. 8KB = 8,192 bytes

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Software